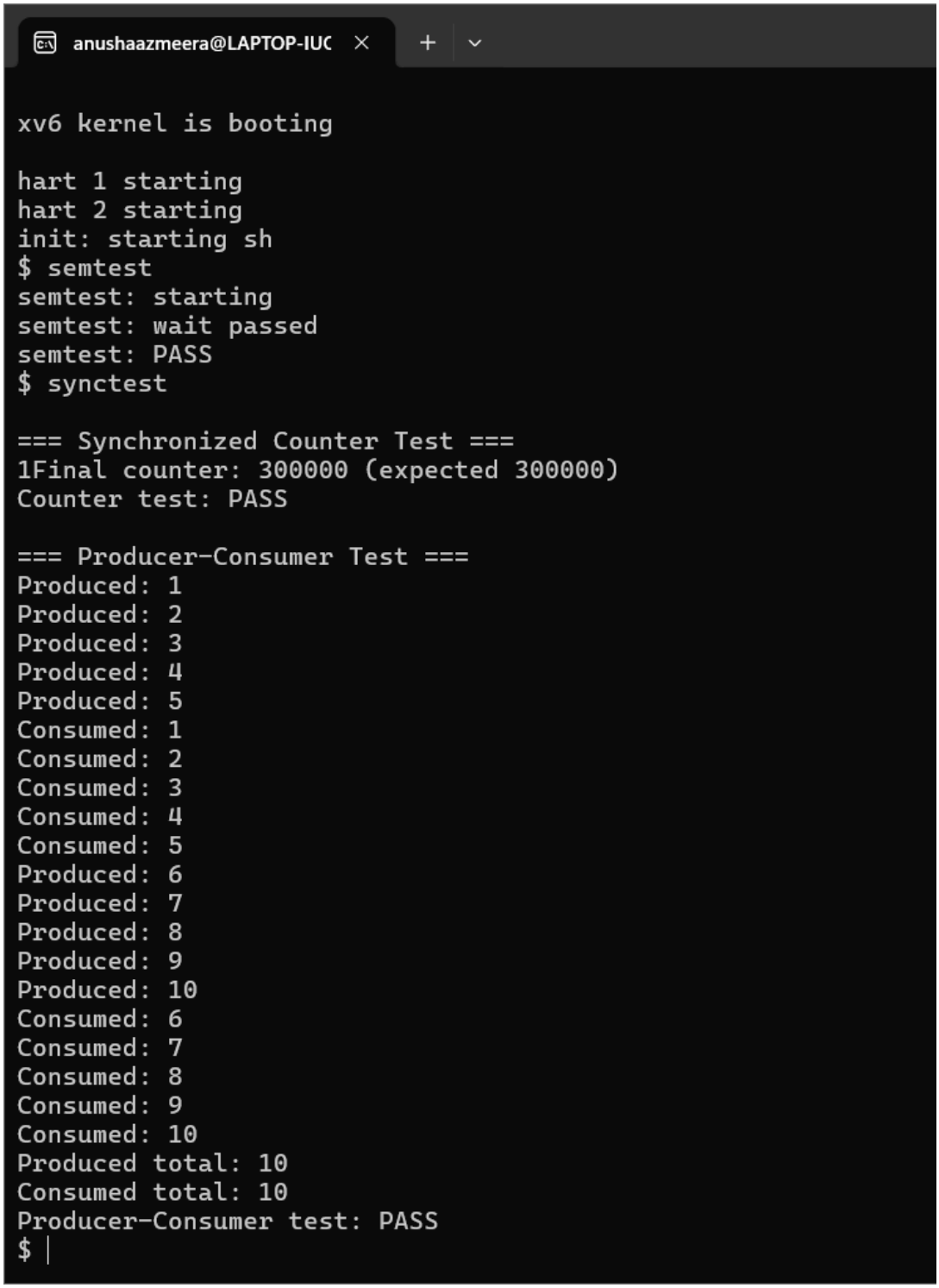
Q1:

```
smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,i
f=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-b
us.0

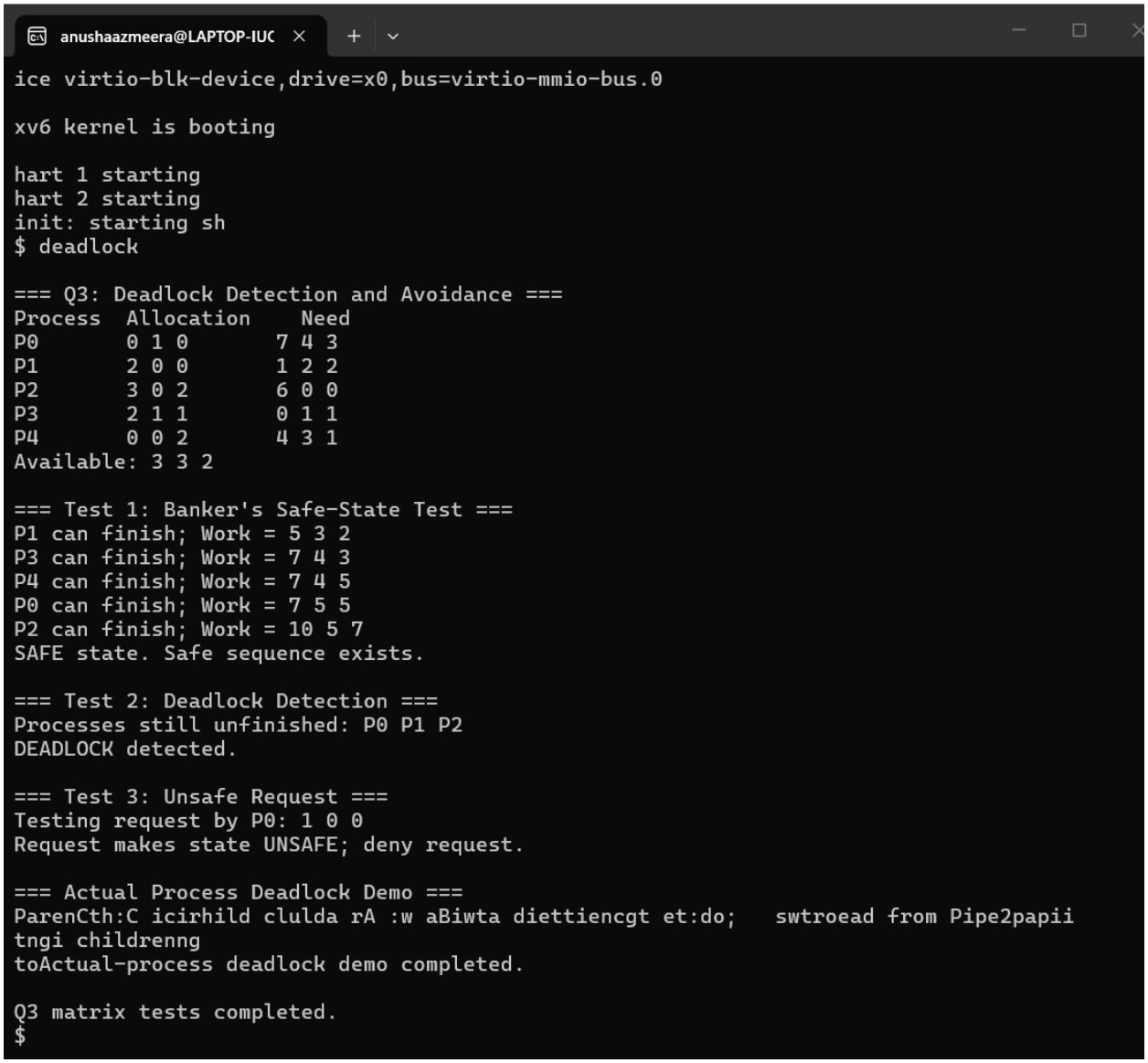
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ threadtest
threadtest: creating 3 threads
tthtrhread 1e afdi h2r nfiieshed
nished
ad 3 finished
threadtest: counter = 203917; expected 300000
threadtest: a value below 300000 demonstrates the Q1 race condition
$ QEMU: Terminated
anushaazmeera@LAPTOP-IUOJJ34L:~/ASSIGN7_2401mc05/code/xv6-riscv$ cd ~/ASSIGN
```

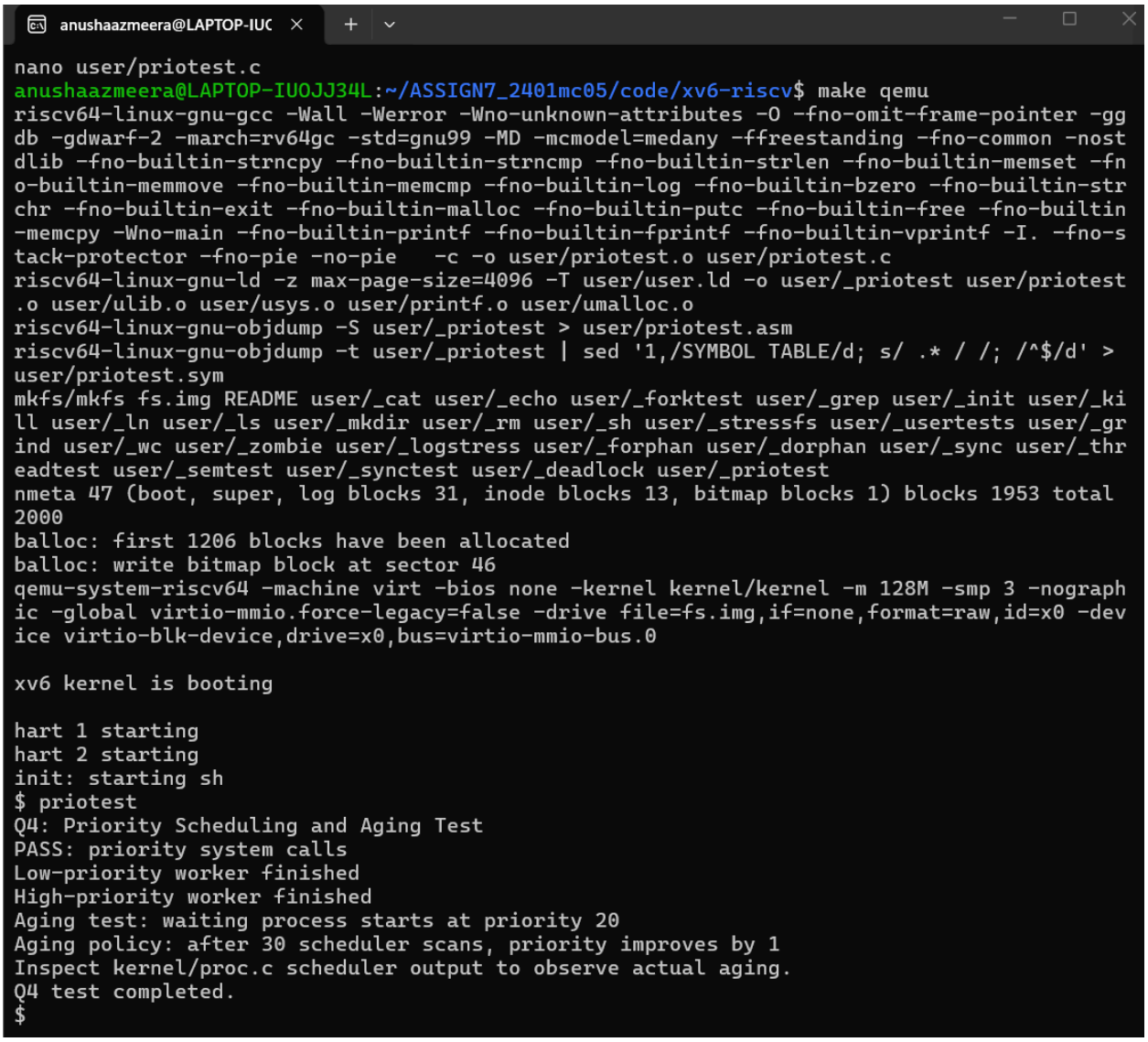
Q2:


anushaazmeera@LAPTOP-IUC
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ semtest
semtest: starting
semtest: wait passed
semtest: PASS
$ synctest
=== Synchronized Counter Test ===
1Final counter: 300000 (expected 300000)
Counter test: PASS
=== Producer-Consumer Test ===
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Produced: 5
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
Produced: 6
Produced: 7
Produced: 8
Produced: 9
Produced: 10
Consumed: 6
Consumed: 7
Consumed: 8
Consumed: 9
Consumed: 10
Produced total: 10
Consumed total: 10
Producer-Consumer test: PASS
$

Q3


anushaazmeera@LAPTOP-IUC
ice virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ deadlock
=== Q3: Deadlock Detection and Avoidance ===
Process Allocation Need
P0 0 1 0 7 4 3
P1 2 0 0 1 2 2
P2 3 0 2 6 0 0
P3 2 1 1 0 1 1
P4 0 0 2 4 3 1
Available: 3 3 2
=== Test 1: Banker's Safe-State Test ===
P1 can finish; Work = 5 3 2
P3 can finish; Work = 7 4 3
P4 can finish; Work = 7 4 5
P0 can finish; Work = 7 5 5
P2 can finish; Work = 10 5 7
SAFE state. Safe sequence exists.
=== Test 2: Deadlock Detection ===
Processes still unfinished: P0 P1 P2
DEADLOCK detected.
=== Test 3: Unsafe Request ===
Testing request by P0: 1 0 0
Request makes state UNSAFE; deny request.
=== Actual Process Deadlock Demo ===
ParenCth:C icirhild clulda rA :w aBiwta diettiencgt et:do; swtroead from Pipe2papii
tngi childrenng
toActual-process deadlock demo completed.
Q3 matrix tests completed.
$

Q4 :


anushaazmeera@LAPTOP-IUC
nano user/priotest.c
anushaazmeera@LAPTOP-IUOJJ34L:~/ASSIGN7_2401mc05/code/xv6-riscv$ make qemu
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -gg
db -gdwarf-2 -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nost
dlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fn
o-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-str
chr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin
-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-s
tack-protector -fno-pie -no-pie   -c -o user/priotest.o user/priotest.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_priotest user/priotest
.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_priotest > user/priotest.asm
riscv64-linux-gnu-objdump -t user/_priotest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' >
user/priotest.sym
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_ki
ll user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_gr
ind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_thr
eadtest user/_semtest user/_synctest user/_deadlock user/_priotest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total
2000
balloc: first 1206 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nograph
ic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -dev
ice virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ priotest
Q4: Priority Scheduling and Aging Test
PASS: priority system calls
Low-priority worker finished
High-priority worker finished
Aging test: waiting process starts at priority 20
Aging policy: after 30 scheduler scans, priority improves by 1
Inspect kernel/proc.c scheduler output to observe actual aging.
Q4 test completed.
$